Reflection

Camille: My contribution focused on sections 7, 8, and 9, including log normalization, MinMax and Standard scaling, and SMOTE class balancing. Comparing the data before and after these steps helped me understand how preprocessing changes feature distributions and ranges. The imbalanced-data demonstration also showed me why class balance matters when preparing to predict customer churn. One of my biggest takeaways was the importance of fitting transformations on training data only, then applying them to the sealed test set without introducing data leakage. Working with my team helped me connect my part to the complete pipeline and feel more confident using Python for data preparation.

Chris: I learned much about exploratory data analysis and problem definition for machine-learning in this assignment. The analysis revealed several relevant patterns to use for predicting customer churn. First, it is essential to inspect the data before building models for analysis. There are many ways in which data can be of very poor quality (unrealistic values, far too many features that are largely redundant, etc.). Selecting evaluation metrics for a model is not simply a matter of choosing the one that reports highest accuracy. For a churn-prediction system, for example, recall and the associated F1-score are of particular interest because there is nothing worse than a model predicting that customers will not churn who in fact do not and then lose all their business. Interpreting correlations between customer-engagement features and churn is challenging without suspecting causality. This assignment thus connected the stages of the workflow for machine-learning that I had not yet implemented in detail and reminded me of the importance of documenting data limitations when analyzing results from a model.

The analysis of the redundant features was completed and how the findings from the EDA could be used to inform the preprocessing of the data was documented. In Section 11, the reasons why churn prediction is a supervised binary-classification problem are explained, and models that will be used for the final project are proposed. The evaluation metrics that will be used are precision, recall, F1 score, the confusion matrix and the ROC-AUC plot. Limitations to the use of this data for a machine-learning experiment are that the data is simulated, and it does not contain any time information. Much can be learned from completing an EDA on a data set and how it can inform the preprocessing of the data for a machine-learning experiment. Also, it is very important to complete all analysis on the training data to avoid data leakage.

Rigoberto: During this project, I learned how important data preprocessing and feature engineering are when preparing data for machine learning. I gained a better understanding of the difference between ordinal encoding and one-hot encoding and why it is important to choose the correct method for different

types of categorical variables. I also learned that creating new features can help reveal useful patterns in customer behavior that may not be obvious in the original dataset.

One challenge was understanding which features should be kept, transformed, or removed to avoid unnecessary or redundant information. Working on this project helped me improve my Python and data analysis skills while learning how to use pandas, numerical transformations, and data visualization. Overall, this project gave me a better understanding of how raw data can be prepared for a machine learning model to predict customer churn. I can use these skills in future projects involving data analysis and artificial intelligence.

Min Ma Kha

This project has changed my view of the gap between raw data and model ready data, a switch that was done mainly by discrediting what I thought of the Netflix dataset to start with. At first, the dataset appeared flawless of 5000 full rows, no missing values, no duplicates, a target that was divided well and the data processing seemed like a matter of putting a tick in some boxes. When I dig deeper the dataset contained daily views above 24 hours, a price column that mimicked the plan group, and distributions that were so badly skewed that it could interfere with any distance-based model but could not have shown up via the isnull() function. On top of that, the most valuable lesson was the approach towards the order of work steps: at first, separating the data into 70/30 groups was a formality, but only until I realized that the median derived from all 5000 rows assigns the test data to the training dataset and impacts the result in a way that cannot be corrected later on. Additionally, I was able to start seeing preprocessing as a sequential process. The similar demographic make-up and almost perfect balance of classes implies that the dataset is simulated, which lessens the validity of the findings. However, this shows that the fundamental lesson is to question data rather than take it at face value.